ADSP 31014 Statistical Models for Data Science
Homework 4

(5 points) Question 1a
Mean Estimator: Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ be an i.i.d. sample of n observed data points. In our lecture, we noted the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is an unbiased estimator for the unknown parameter μ. Let's re-run the R code simulation in the lecture as follows to verify:

```
num.of.estimate <- 1000000
sample.means <- numeric(num.of.estimate)

for (i in seq_len(num.of.estimate)) {
  n <- 10
  x <- rnorm(n, mean = 7, sd = 2)
  sample.means[i] <- mean(x)
}

mean(sample.means)
```

(5 points) Question 1b
Median Estimator: Based on the same scenario in Question 1a, since μ is also the median i.e. 50%-tile of the probability distribution, we might consider using the sample median $X_n^{50\%}$ of the observed data to estimate μ. Let's do a simulation to find out if the sample median is also an unbiased estimator for μ or not. Write some R code as follows:

```
sample.medians <- numeric(num.of.estimate)

for (i in seq_len(num.of.estimate)) {
  n <- 10
  x <- rnorm(n, mean = 7, sd = 2)
  ... # Compute median
}

...
```

Based on your simulation result, is the sample median unbiased for μ?

(5 points) Question 1c
Mid-Range Estimator: Based on the same scenario in Question 1a, the sample mid-range is

$$X_n^{\mathrm{MR}} = \frac{\left(\min_{i=1,\ldots,n} X_i\right) + \left(\max_{i=1,\ldots,n} X_i\right)}{2}$$

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Since the normal distribution is symmetric around μ, X_n^{MR} might also be a reasonable estimator for μ. Is it unbiased? Do a simulation similar to Question 1a to find out.

(5 points) Question 1d

UMVUE: Besides unbiasedness**,** having low variance (across different data samples) is another desirable characteristic of an estimator. Continuing from Question 1a-c, let's compute

```
var(sample.means)
...
```

Based on your result, which one among the sample mean, median, and mid-range is the best estimator for μ? See https://en.wikipedia.org/wiki/Minimum-variance_unbiased_estimator for more information.

(15 points) Question 2

Unbiased Estimate of Noise σ^2: Let $r_1, \ldots, r_n$ be the n residuals of a linear regression model. In our lecture, we noted that both of the following ways

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} r_i^2 \quad \text{or} \quad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n} r_i^2$$

to estimate the variance parameter σ^2 of the noise term are biased. The unbiased way to estimate σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n-p}\sum_{i=1}^{n} r_i^2$$

where $p = \dim(\vec{\beta})$ is the total number of intercept and slope parameters i.e. model degrees of freedom. Let's run some simulation to verify this.

```
num.of.repeat <- 10000
sigma.sq.hat <- numeric(num.of.repeat)

for (i in seq_len(num.of.repeat)) {
  # Step 1: Simulate design matrix X
  n <- 100
  d <- 5
  X <- cbind(1, matrix(rnorm(n * d), n, d))

  # Step 2: Simulate response y
  beta <- 1:(d + 1) # True parameter
  sigma.sq <- 0.09 # True parameter
  y <- X %*% beta + rnorm(n, sd = sqrt(sigma.sq))

  # Step 3: Fit linear regression model
  data <- data.frame(X, y)
  model <- lm(y ~ ., data = data)

  # Step 4: Compute variance of residuals
  sigma.sq.hat[i] <- var(residuals(model)) # Based on 1 / (n - 1) * sum(...)
}
```

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```
mean(sigma.sq.hat)

p <- length(beta)
mean(sigma.sq.hat * (...)) # Unbiased for sigma.sq
```

In your simulation, what is the value of `mean(sigma.sq.hat)`? Does this imply Step 4 tend to under-estimate or over-estimate the true value of the σ^2 parameter? What is the multiplicative adjustment `(...)` necessary to make it unbiased for σ^2?

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